



Optimization of Hybrid Renewable Energy Systems for Sustainable Rural Electrification in Kwara State, Nigeria: A Techno-Economic and Environmental Assessment

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Authors' contributions

This work was carried out in collaboration among all authors. Author NAO carried out the research. Author JAO wrote the protocol, and wrote the first draft of the manuscript. Author OAA managed the literature searches. Author MOO supervise the research. All authors read and approved the final manuscript.

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ABSTRACT

Access to reliable and affordable electricity remains a major challenge for rural communities in Nigeria, particularly in Kwara State, where weak grid infrastructure and reliance on costly fossil fuels constrain socio-economic development. Despite abundant renewable resources such as solar, wind, hydro, and biomass, these remain underutilized. This study designed and optimized Hybrid Renewable Energy Systems (HRES) for three off-grid communities—Owode, Abe-Eya, and Ila-Oja Aboto—using a mixed-method approach that combined household energy surveys, field-based resource assessments, and techno-economic simulations. HOMER Pro (v3.14.2) was used to develop optimal hybrid configurations integrating solar photovoltaic (PV), wind, and biomass systems, while RETScreen Expert evaluated financial and environmental performance. Results showed strong solar irradiance (~5.85 kWh/m²/day), moderate wind speeds (3.8–4.2 m/s) in Abe-Eya, and abundant biomass availability (1.5–2.2 tons per household per year) in Owode and Ila-Oja Aboto. The optimal configuration included 54 × 215 W PV modules, 2 × 1 kW wind turbines, 4 × 1 kW batteries, and 34 kVA converters, with installation costs of ₦22.32 million (Owode), ₦24.48 million (Abe-Eya), and ₦24.96 million (Ila-Oja Aboto), generating 25,754–31,643 kWh annually. Economic analysis indicated strong viability with NPV > ₦550 million, IRR of 142–158%, payback periods of 3.2–3.5 years, and benefit–cost ratios of 1.15–1.17. Greenhouse gas emissions decreased by 63–70%, from 6,000–7,500 kg CO₂/year to 613–959 kg CO₂/year. Sensitivity analysis confirmed robustness under ±25% cost variation. The study demonstrates that community-specific solar-biomass and solar-wind hybrid systems provide reliable, cost-effective, and sustainable electrification pathways for rural Nigeria, advancing the national energy transition and Sustainable Development Goal 7.

Keywords: Hybrid renewable energy systems (HRES); rural electrification; techno-economic optimization; renewable energy integration; sustainable development.

1. INTRODUCTION

1.1 Background of the Study

Nigeria, the most populous nation in Africa with a population over 235 million, possesses a wealth of both renewable and non-renewable energy resources. Despite this considerable potential, the nation continues to face a significant electricity supply issue, with less than fifty percent of the population having reliable access to electricity (Fasina et al., 2020). The primary renewable resources accessible are sun, biomass, wind, and hydropower, each of which can substantially address the nation's increasing energy requirements. The use of these resources is regrettably limited, resulting in the energy system's inability to satisfy demand and hindering socio-economic development (Oyedokun et al., 2022; Bamisile et al., 2022).

The issue of power availability is especially severe in rural regions where infrastructure is inadequate, population density is low, and the expenses associated with grid expansion are exorbitant. In Kwara State, located in north-central Nigeria, numerous rural and semi-urban villages endure ongoing energy poverty. Despite grid extension being the national strategy for

electrification, the cost and logistical challenges of extending transmission lines to rural villages render it impracticable (Oyedokun et al., 2023). As a result, rural homes and small businesses rely on expensive and unsustainable options like diesel or petrol generators and kerosene lamps, which are ecologically harmful and inadequate for contemporary energy requirements. The International Energy Agency (2021). The International Energy Agency (2021) predicts that more than 600 million individuals in Sub-Saharan Africa lack access to electricity, with many relying on harmful energy sources.

Hybrid Renewable Energy Systems (HRES) have emerged as a possible answer to these difficulties. By integrating two or more renewable resources such as solar, wind, hydro, and biomass alongside storage technologies, HRES can supply reliable, continuous, and sustainable electricity. These systems reduce the intermittency associated with single renewable technologies, lower dependence on fossil fuels, and promote energy security (Ariyo et al., 2020). Importantly, HRES disperse investment among several energy sources, giving cost-effective, environmentally sustainable, and socially acceptable energy solutions suitable for rural areas (Chendo, 2022).

1.2 Energy and Electricity Situation in Nigeria

Despite being endowed with fossil and renewable resources, Nigeria continues to have one of the lowest per capita energy access rates internationally. Only 55.4% of the entire population and roughly 24.6% of the rural population are connected to the grid (The World Bank, 2023). The energy supply chain suffers from inadequate generation capacity, weak transmission infrastructure, and high distribution losses, leading to frequent outages and unstable supply (Araoye et al., 2018). Although capacity has risen in recent years, most upgrades have favored metropolitan centers, leaving rural regions neglected. Nigeria's national grid is dominated by natural gas and hydropower, while solar and biomass make up a tiny share of supply. Heavy reliance on fossil fuels is unsustainable due to price volatility, resource depletion, and environmental implications (Fasina et al., 2023). In contrast, renewable resources provide tremendous potential: solar radiation averages 5–7 kWh/m²/day across the country, while biomass wastes from agriculture are abundant and unused. However, grid development remains capital demanding, requiring investments that Nigeria struggles to organize. This fact underscores the necessity for decentralized solutions such as microgrids and HRES that can offer reliable, off-grid electricity in rural communities.

1.3 Rural Electrification in Nigeria

Rural electrification remains one of Nigeria's most serious development concerns. Less than 30% of rural families have dependable electrical connections, and many areas are fully off-grid (Fasina et al., 2023). Energy poverty disproportionately affects women and vulnerable groups who depend on traditional biomass for cooking, heating, and lighting, leading to severe health concerns and perpetuating cycles of poverty. Moreover, the lack of consistent power constrains agricultural mechanization, irrigation, agro-processing, healthcare delivery, and education, driving rural–urban migration.

Government efforts to encourage rural electricity have been modest. Institutions such as the Rural Electrification Board (REB) have begun initiatives; however, challenges such as limited finance, poor infrastructure, vandalism, and lack of qualified labor have hampered sustainability

(Lal et al., 2021). As a result, rural electrification efforts must look beyond conventional grid extension to decentralized renewable energy options. Globally, decentralized systems powered by renewables are now widely recognized as cost-effective, scalable, and sustainable choices for rural areas.

1.4 Profile of Kwara State and Energy Potentials

Kwara State, located in north-central Nigeria, encompasses 33,773 km² and has an estimated population of 3.2 million (NPC, 2016). The state is largely agrarian, with farming, small-scale industry, and trading as the main economic activity. Its meteorological characteristics, governed by the Inter-Tropical Convergence Zone, generate an average annual rainfall of 1700 mm and high solar radiation, making it favorable for renewable energy deployment (Fasina et al., 2023). Despite this promise, electricity remains a big barrier, particularly in rural populations. Many households and companies rely extensively on diesel generators, which increase expenses, harm the environment, and hamper sustainable development. Although programs such as the "Light Up Kwara" project have tried to boost renewable energy adoption, implementation has been delayed, and many rural villages remain neglected.

1.5 Renewable and Non-Renewable Energy Resources in Nigeria

Nigeria's energy mix is still dominated by crude oil, natural gas, and coal, which together generate over 80% of electrical generation (Lal et al., 2021). While hydropower produces roughly 38.5% of grid electricity, minor hydro resources remain mostly unexploited despite the potential to surpass 3,500 MW (Oyedokun et al., 2022). Solar energy offers the most promise, with theoretical annual potential surpassing 1.8×10^{15} kWh (Ariyo et al., 2020). Yet adoption has been limited to small-scale applications. Wind speeds average 2–5 m/s in most places, whereas agricultural biomass and urban garbage might create millions of cubic meters of clean energy every day (Araoye et al., 2018). In contrast, prolonged dependency on fossil fuels entails hazards of carbon emissions, resource depletion, and global price shocks (Energy Commission of Nigeria, 2020). Transitioning towards renewable energy is consequently vital to Nigeria's long-term energy security and climate change obligations.

1.6 Hybrid Renewable Energy Systems and Optimization

HRES integrates different renewable energy resources frequently backed with storage or backup systems to provide dependable and steady power supply in contexts where grid extension is not viable (Abdullahi et al., 2020). Solar–biomass and solar–wind hybrids are particularly successful in rural environments, as they combine complementary resources to assure constant supply. Optimization plays a major part in HRES design. Traditional trial-and-error procedures often lead to overdesign or inefficiency. By contrast, optimization methods like genetic algorithms, simulated annealing, and particle swarm optimization systematically discover the best cost-effective mix of resources, storage, and backup capacity (Adepoju et al., 2022; Sinha & Chandel, 2023; Khan & Iqbal, 2021; Tijani et al., n.d.; Odou et al., 2020). Commercial tools like HOMER Pro give sophisticated platforms to simulate, optimize, and evaluate alternative hybrid configurations, while RETScreen offers thorough evaluation of life-cycle costs, environmental consequences, and greenhouse gas (GHG) emission reductions.

Studies across Sub-Saharan Africa demonstrate that improved HRES may produce energy at leveled costs competitive with diesel generators while decreasing GHG emissions by over 70% (Koutroulis et al., 2006). Yet in Nigeria, and Kwara State especially, deployment of such techniques remains limited due to financial hurdles, lack of demand for resource data, and insufficient governmental support.

The launch underlines Nigeria's chronic energy access issues, particularly in rural areas where electrification rates remain low. Kwara State, despite its vast renewable resource base, has seen little implementation of HRES. Existing studies indicate the technical feasibility and sustainability of optimized hybrid systems internationally, but empirical evidence from Nigeria remains rare. This work tries to solve that gap by employing HOMER Pro for hybrid system modeling and optimization and RETScreen for economic and environmental evaluation. Focusing on three rural towns in Kwara State Owode, Ila-Oja Aboto, and Abe-Eya the research intends to demonstrate that correctly constructed HRES can offer reliable, inexpensive, and clean electricity for rural homes and enterprises. By aligning with Nigeria's energy access targets and global climate commitments, our work adds to

scalable frameworks for rural electrification in Sub-Saharan Africa.

2. MATERIAL AND METHODS

This work adopts a mixed-method research strategy, combining primary data collecting, secondary data gathering, and advanced modeling tools to develop and optimize a hybrid renewable energy system (HRES) for rural electrification in Kwara State, Nigeria. Primary data were acquired through standardized questionnaires delivered to families and small companies to examine energy demand patterns, appliance usage, and willingness to embrace renewable energy solutions. Physical site inspections were done to examine land use, accessibility, and availability of renewable resources. Secondary data, including solar irradiance, wind speed, temperature, and clearness index, were collected from the National Aeronautics and Space Administration (NASA) database. Together, these records supplied the baseline for resource evaluation and load demand modeling.

Three rural communities were selected, each representing one of the three senatorial districts of Kwara state: Owode (Kwara South), Ila-Oja Aboto (Kwara Central), and Abe-Eya (Kwara North). These places were chosen for their agricultural relevance, socio-economic activities, and lack of connection to grid electricity. Field surveys included load estimation for households, small-scale enterprises, and public facilities such as schools and health clinics. The survey also documented locally available biomass feedstock, such as crop residues, animal waste, and organic garbage, to estimate their potential for energy generation. The inclusion of varied sites enabled a comparative evaluation and ensured representativeness of rural energy concerns across the state.

Resource assessment for sun, wind, and biomass was accomplished using a combination of meteorological data analysis and field assessments. Solar potential was analyzed using monthly average global horizontal irradiance (GHI) and clearness index, whereas wind potential was derived from site-specific wind speed distributions at 10 m height. Biomass availability was estimated using household-level agricultural residue quantification and calculation of livestock waste suitable for biogas production. These characteristics were integrated into the hybrid optimization of multiple energy resources

(HOMER Pro) software to model various hybrid configurations. The program was used to optimize system components solar PV, wind turbines, biogas systems, storage units, and converters based on cost, technical reliability, and resource availability.

For economic and environmental evaluation, the RETScreen software was applied to analyze key performance measures, including net present cost (NPC), levelized cost of electricity (LCoE), internal rate of return (IRR), and payback period. The tool also gave estimates of greenhouse gas (GHG) emission reductions relative to baseline diesel-powered alternatives. This dual-software method ensured both technical optimization and feasibility assessment of the proposed HRES. By merging household-level demand surveys, resource assessments, and simulation-based optimization, the methodology creates a strong framework for constructing replicable rural electrification models in Nigeria and other Sub-Saharan African contexts.

3. RESULTS AND DISCUSSION

3.1 Feasibility Assessment Results

This study investigated the feasibility of establishing Hybrid Renewable Energy Systems (HRES) in three strategically selected off-grid rural communities in Kwara State, Nigeria: Owode (8°22'N, 5°20'E), Ila-Oja Aboto (4°55'N, 8°15'E), and Abe-Eya (9.0719°N, 4.5075°E). These sites represent the three senatorial districts and were chosen owing to their lack of grid connectivity, renewable resource availability, and diverse energy demand characteristics. The feasibility evaluation was based on expected

load profiles, renewable energy potentials (solar, wind, and biomass), and local environmental conditions. The three villages have similar demand patterns dominated by domestic and agricultural applications, including lights, mobile phone charging, radio, television, refrigeration, and small appliances. As demonstrated in Figs. 1–3, load demand continuously peaks between 7:00 PM and 10:00 PM due to nighttime household activities, decreases steeply between 11:00 PM and 6:00 AM, and exhibits a minor comeback between 7:00 AM and 10:00 AM before stabilizing at low levels during farming hours (11:00 AM–6:00 PM).

The matching of load demand with renewable energy availability improves the rationale for hybrid systems. Abe-Eya offers tremendous potential for a solar-wind hybrid system, with average wind speeds of 3.8–4.2 m/s complementing high solar irradiation. In contrast, Owode and Ila-Oja Aboto are better suited for solar-biomass hybrids, utilizing consistent solar resources and ample agricultural residues, animal waste, and home organics estimated at 1.5–2.2 tons per household annually.

Average solar exposure across the three locations is roughly 5.85 kWh/m²/day, closely coinciding with nighttime and daytime loads. High clearness indices and moderate ambient temperatures (25–34°C) boost photovoltaic (PV) efficiency and durability. Wind at Abe-Eya provides extra electricity throughout early morning and nighttime hours when solar output drops. Furthermore, community surveys verified ample rooftop space on schools, health centers, and marketplaces (≈230 m² per village) for PV installation, avoiding competition with farming.

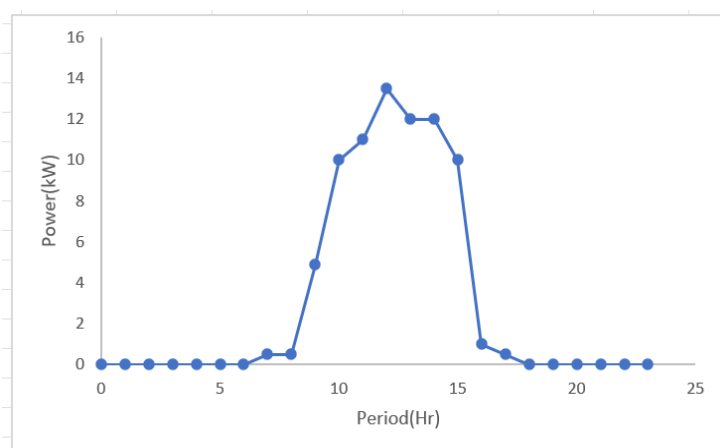


Fig. 1a. Hourly Load Demand Profile of Owode

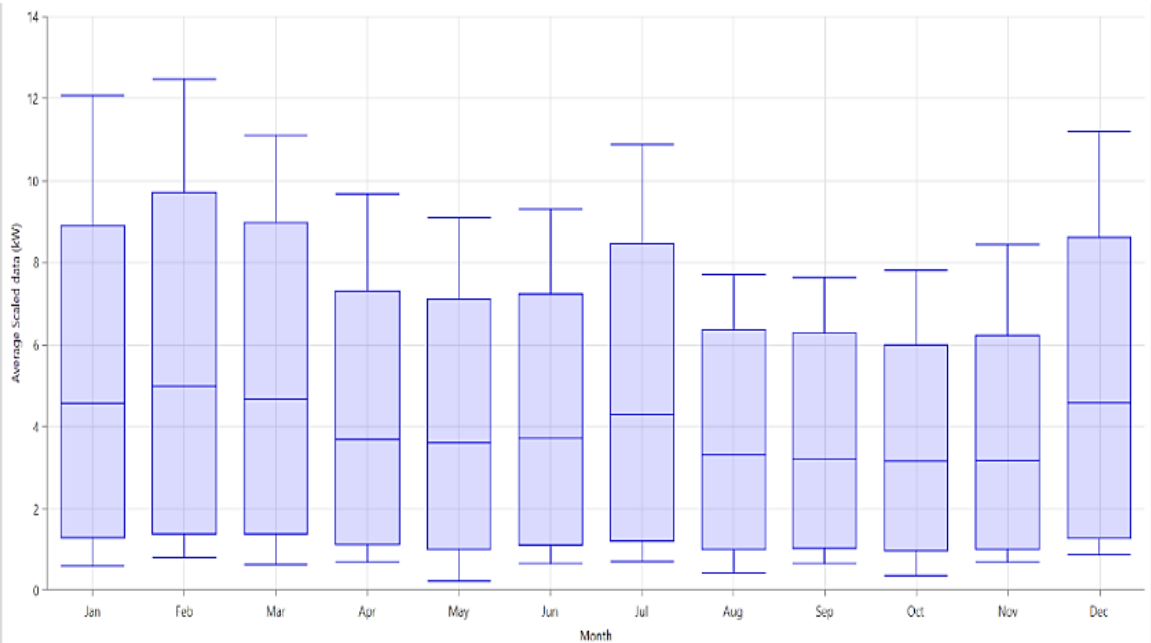


Fig. 1b. Seasonal Load Demand Profile of Owode

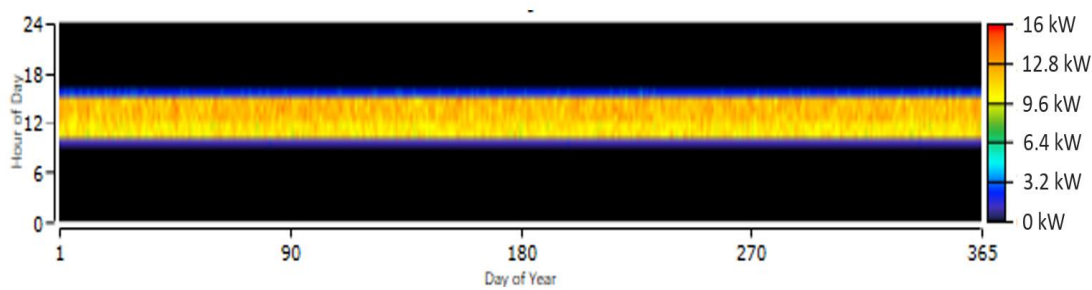


Fig. 1c. Yearly Load Demand Profile of Owode

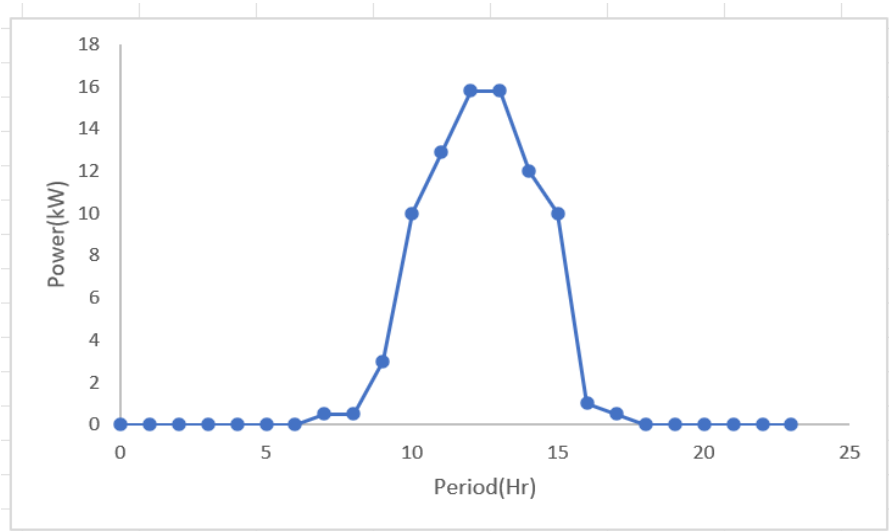


Fig. 2a. Hourly Load Demand Profile of Abe-Eya

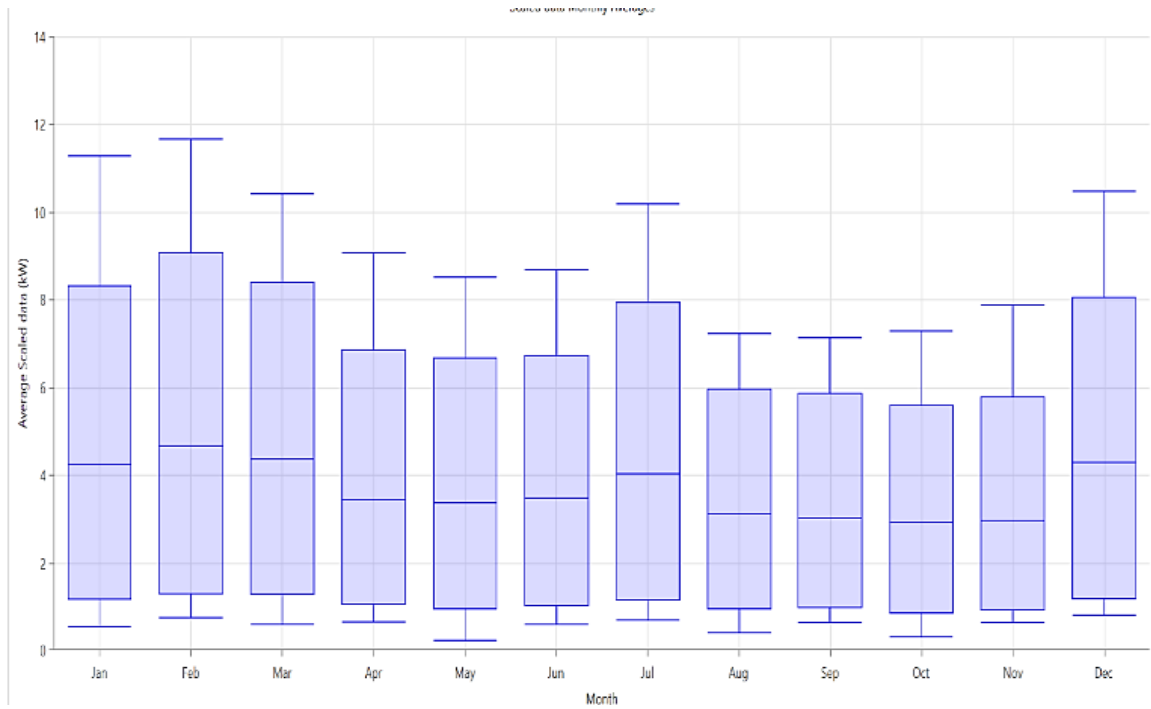


Fig. 2b. Seasonal Load Demand Profile of Abe-Eya

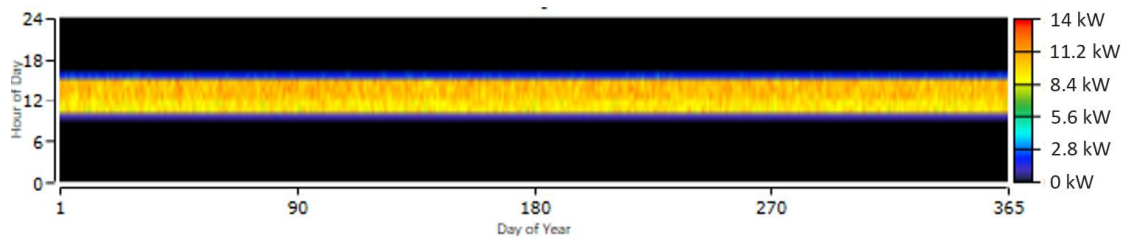


Fig. 2c. Seasonal Load Demand Profile of the Abe-Eya

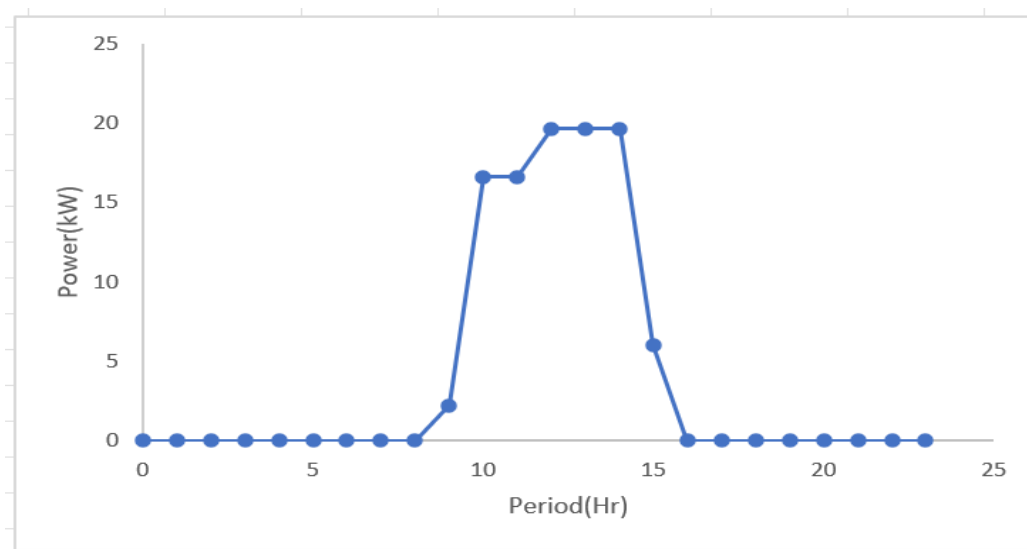


Fig. 3a. Hourly Load Demand Profile of Ila-Oja Aboto

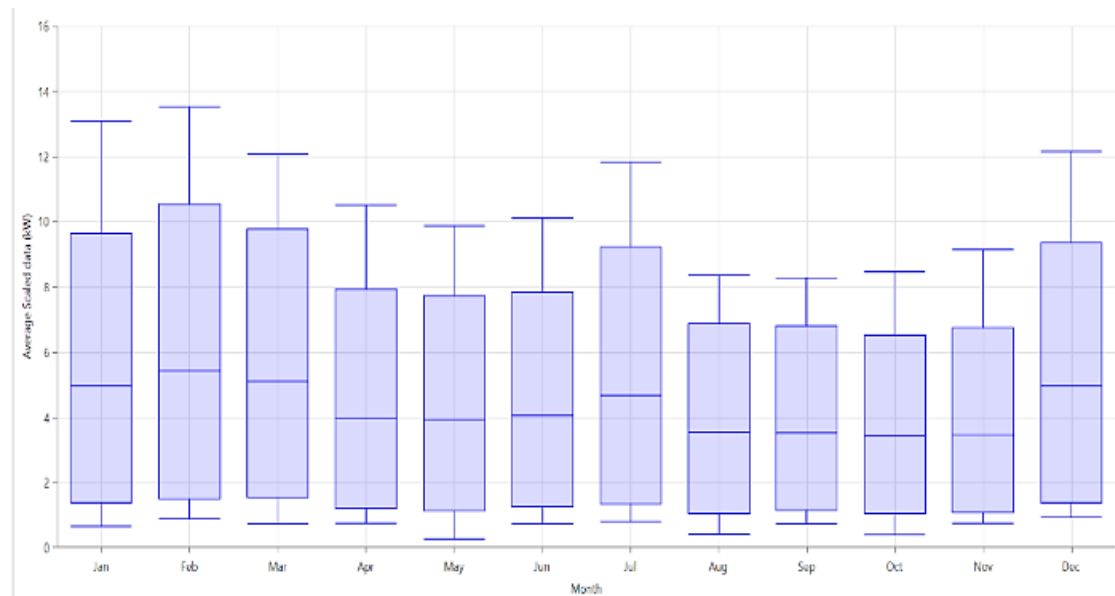


Fig. 3b. Seasonal Load Demand Profile of Ila-Oja Aboto

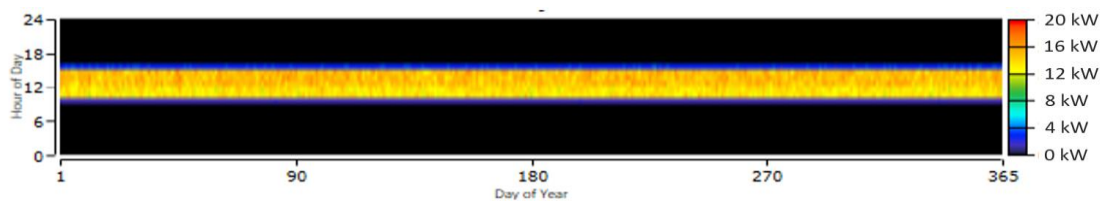


Fig. 3c. Yealy Load Demand Profile of the Ila-Oja Aboto

Overall, the analysis indicates the technological and spatial viability of deploying HRES in all three communities. Each area has distinct capabilities that justify different renewable resource combinations, delivering reliable, site-specific energy solutions capable of increasing rural electrification, raising living standards, and promoting socio-economic growth.

3.2 Optimized Hybrid Renewable Energy System

The hybrid systems were optimized in HOMER Pro (v3.14.2) considering solar PV panels, wind turbines, biomass generators, battery storage, and converters. Optimal component sizes are

summarized in Tables 1–3. For Abe-Eya, the HOMER simulation recommends 54 units of 215 W PV modules, two 1 kW wind turbines, four 1 kW lead-acid batteries, and a 34 kVA converter. Owode and Ila-Oja Aboto, optimized as solar-biomass hybrids, require the same amount of PV modules and batteries, supported by one 5 kW biomass gasifier generator each.

In In Abe-Eya, the PV array is expected to operate for ~4,313 hours yearly, far exceeding the community's requirement (~2,555 hours). The wind turbine, albeit modest in energy production, maintains supply stability during low-solar periods and minimizes battery stress, as depicted in Fig. 4. In Owode and Ila-Oja Aboto,

Table 1. Optimal sizes of the main components of Owode

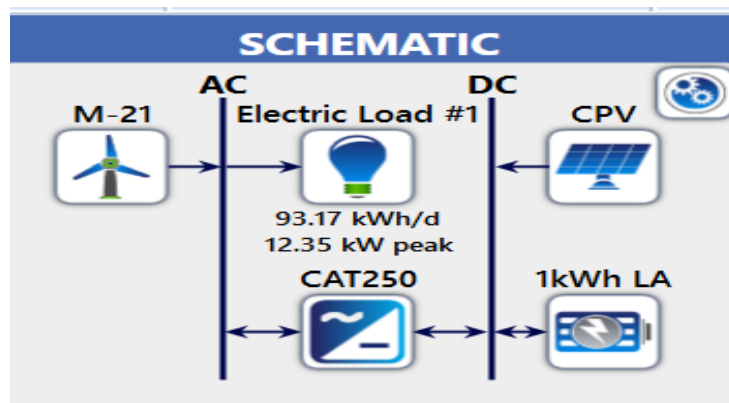
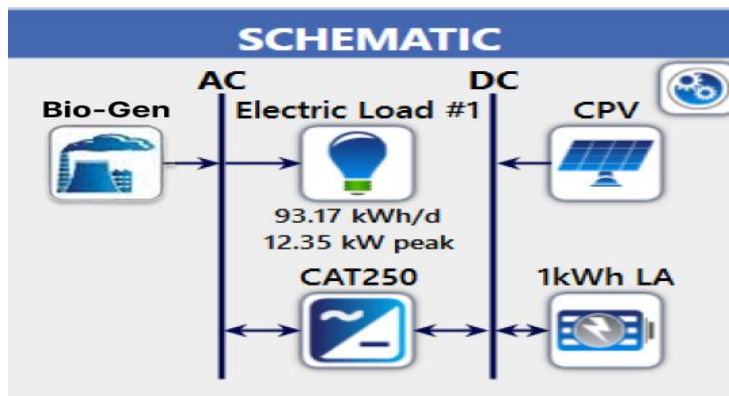
Component	Specification	Quantity
Solar PV Panel	Sunpower215PL-SUNP-SNR-215 (215 W)	54 units
Wind Turbine	Bergey Excel 1 (1 Kw, cut in speed 3.5m/s)	2 units
Battery	Generic 1kW Lead Acid	4 units
Power Converter	Generic system converter (34 kVA)	1 unit
Charge Controller	MTTP Hybrid Controller (solar/wind compatible)	1 unit

Table 2. Optimal sizes of the main components of Abe-Eya

Component	Specification	Quantity
Solar PV Panel	Sunpower215PL-SUNP-SNR-215 (215 W)	54 units
Biomass Generator	5 KW Biomass Gasifier Generator (using agricultural residue)	1 unit
Battery	Generic 1kW Lead Acid	4 units
Power Converter	Generic system converter (34 kVA)	1 unit
Charge Controller	MTTP Hybrid Controller (solar/wind compatible)	1 unit

Table 3. Optimal sizes of the main components of Ila-Oja Aboto

Component	Specification	Quantity
Solar PV Panel	Sunpower215PL-SUNP-SNR-215 (215 W)	54 units
Biomass Generator	5 KW Biomass Gasifier Generator (using agricultural residue)	1 unit
Battery	Generic 1kW Lead Acid	4 units
Power Converter	Generic system converter (34 kVA)	1 unit
Charge Controller	MTTP Hybrid Controller (solar/wind compatible)	1 unit

**Fig. 4. Schematic Diagram of the Proposed Hybrid Renewable Energy System in Abe-Eya****Fig. 5. Schematic Diagram of the Proposed hybrid renewable energy system in Owode and Ila-Oja Aboto**

solar generation ranges from 25,754 to 31,643 kWh/year, constantly surpassing local demand. Surplus electricity may support agro-processing, water pumping, or other profitable uses as indicated in Fig. 5. Biomass generators provide stable backup during extended cloudy

circumstances, ensuring resilience without fossil fuel reliance.

In all locations, Battery Energy Storage Systems (BESS) manage short-term swings and secure supply during peak demand periods. This

personalized method provides reliability, scalability, and cost-effectiveness, while drastically lowering dependence on diesel generators and associated greenhouse gas (GHG) emissions.

3.3 Comprehensive Analysis of the Proposed System

Financial Analysis: RETScreen software was used to evaluate financial viability. Key assumptions included a discount rate of 9%, an inflation rate of 34.2% (current Nigerian figure), a 20-year project life, and 70% debt financing at 7% interest as shown in Table 4.

Table 4. Financial parameters of the proposed design

Parameters	Values
Fuel cost escalation rate	25%
Inflation rate	34.2%
Discount rate	9%
Reinvestment rate	9%
Project life	20
Debt ratio	70%
Debt interest rate	7%
Debt payment	10

Initial investment requirements are projected at NGN 22.32 million for Owode, NGN 24.48 million for Abe-Eya, and NGN 24.96 million for Ila-Oja Aboto. Annual operating costs, including O&M and debt servicing, mirror these values. Despite high upfront costs, the system produces significant savings: NGN 11.4 million (Owode), NGN 12.1 million (Ila-Oja Aboto), and NGN 11.9 million (Abe-Eya) annually, assuming a Band A electricity rate of NGN 209/kWh. Cumulative cash flow analysis indicates profitability across all sites, with considerable positive inflows beginning in year 2. Replacement of important components (e.g., batteries, inverters) momentarily reduces inflows in certain years, but

long-term trends remain significantly favorable. By year 20, cumulative cash flows exceed NGN 2.1 billion in each community. Financial parameters as indicated in Table 5 affirm excellent viability: Internal Rates of Return (IRR) on equity surpass 140% in all sites; simple payback timeframes average 3.2–3.5 years; and equity payback is less than one year. Net present values are highly positive, and benefit–cost ratios (>115) demonstrate profitability.

3.4 Emission Analysis

The RETScreen emission study demonstrates that the suggested hybrid systems cut GHG emissions by 63–70% compared to baseline fossil fuel usage. Estimated annual emissions are 959.17 kg (Owode), 613.30 kg (Abe-Eya), and 7569.00 kg (Ila-Oja Aboto), compared with >6,000–7,500 kg under traditional systems. These reductions underscore the environmental benefits of hybridization, particularly in rural regions where biomass and PV can directly supplant diesel generators.

3.5 Sensitivity and Risk Analysis

Sensitivity analysis demonstrates that net present value drops when starting costs and grid electricity prices grow. Conversely, reductions in these factors boost NPV and shorten payback periods. Nonetheless, NPVs remain robust even under inflationary scenarios, with minimum values exceeding NGN 550 million across all sites. Monte Carlo-based risk analysis as presented in Table 6 confirmed system resilience to cost and parameter uncertainty. At a 10% risk level, NPVs were between NGN 498 million and NGN 805 million, with equity payback durations between 0.56 and 0.84 years. These results illustrate the strong investment security of the suggested solutions, especially under uncertain economic conditions in Nigeria.

Table 5. Financial viability of the proposed designs

Parameters	Values		
	Owode	Abe-Eya	Ila-Oja Aboto
Internal Rate of Return on Equity	158%	149%	142%
Internal Rate of Return on Asset	61.4%	60.9%	62.1%
Simple Payback Period	3.5 years	3.2 years	3.3 years
Equity Payback	0.79	0.86	0.85
Net Present Value	NGN 554,223,639.10	NGN 554,754,368.90	NGN 554,982, 223.50
Annual Life cycle saving	NGN 80,713,653.07/yr	NGN 78,784,246.33/yr	NGN 79,700,246.00/yr
Benefit-Cost (B-C) ratio	115	117	116

Table 6. Risk analysis report

Parameter	Value	Minimum	Maximum
Initial Costs (NGN)	16,223,297.00	12,167,472.75	20,279,121.25
O&M (NGN)	43,645.00	32,733.75	54,556.25
Fuel cost-proposed case (NGN)	3,525,306.88	2,643,980.16	4,406,633.60
Fuel cost-base case (NGN)	9,642,827.03	7,232,120.27	12,053,533.80
Debit rate (%)	70.00	52.50	87.50
Debit interest rate (%)	7.00	5.25	8.75
Debt term (year)	10.00	7.50	12.50
Net Present Value (NGN)	554,223,639.10	450,737,356.72	657,709,921.48
Equity Payback Period (Years)	0.8150	0.4882	1.2661
Discount Rate (%)	6.00	4.50	7.50
Inflation Rate (%)	13.00	9.75	16.25

4. CONCLUSION

This study investigated and optimized a Hybrid Renewable Energy System (HRES) for off-grid electricity in three rural communities in Kwara State, Nigeria: Owode, Ila-Oja Aboto, and Abe-Eya. The results suggest that personalized hybrid systems, constructed around locally available renewable resources and demand patterns, are both technically feasible and economically viable. With an average daily sun irradiation of 5.85 kWh/m² across all sites, solar energy constitutes the backbone of the system, while moderate wind speeds in Abe-Eya justify solar–wind hybridization, and rich biomass in Owode and Ila-Oja Aboto promotes solar–biomass integration. The optimized arrangements allow 24-hour electrical access, successfully eliminating fossil fuel–based alternatives and decreasing greenhouse gas emissions. By fulfilling important residential and community loads such as lighting, refrigeration, mobile charging, and small appliances, the system fulfills critical rural energy demands while decreasing costs and environmental implications. The findings demonstrate that community-specific HRES can dramatically improve living conditions, education, healthcare, and economic production in underserved places. The successful demonstration in these three villages gives a reproducible paradigm for other rural locations in Nigeria and sub-Saharan Africa. Overall, the study emphasizes the transformative impact of optimized hybrid renewable systems in bridging the rural electrification gap and promoting Nigeria's sustainable development and energy transition agenda.

5. RECOMMENDATION

Based on the results, the following recommendations are recommended for

politicians, development agencies, researchers, and local stakeholders:

- Community-Specific Deployment:** Tailored HRES should be deployed in the examined communities, with solar–biomass systems favored in Owode and Ila-Oja Aboto, and solar wind systems in Abe-Eya. These approaches should be included in Nigeria's broader rural electrification strategy.
- Institutional and Financial Support:** Federal and state governments, in conjunction with private investors and donor organizations, should offer enabling frameworks, including subsidies, tax incentives, and concessional financing, to expedite rural HRES adoption and assure long-term sustainability.
- Capacity Building and Community Engagement:** Training programs should establish local competence in installation, operation, and maintenance, while active community participation should be fostered to enhance ownership and sustainability. Continuous monitoring of solar, wind, and biomass resources is also recommended to enhance performance.
- Replication and Scale-Up:** The demonstrated success in Kwara State should be scaled to other rural communities in Nigeria and across sub-Saharan Africa. Future research should focus on sophisticated storage integration, smart monitoring technologies (IoT, AI), and improved hybrid optimization models to enhance efficiency and resilience.

DISCLAIMER (ARTIFICIAL INTELLIGENCE)

Author(s) hereby declared that NO generative AI technologies such as Large Language Models

(ChatGPT, COPILOT, etc) and text-to-image generators have been used during writing or editing of this manuscript.

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COMPETING INTERESTS

Authors have declared that no competing interests exist.

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